

READYKIT · REQUIREMENTS

ReadyKit requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	RDK-REQ-001	Version	0.3
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/readykit		
Source	docs/03-requirements.md @ 7668a66		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from RDK-CAL-001; R10 and R12 redefined for pip distribution and R11 names readykit.yaml (RDK-DDR-001, D1 and D2); R16 to R18 added (D3, D4, D6); R2 reclassified as not verifiable here; coverage audit checked by seeded faults
0.2	2026-09-25	Amish Chadha	Fifteen measurable requirements for the tool with targets, current status and verification
0.1	2026-09-25	Amish Chadha	Initial scaffold

ReadyKit requirements

ReadyKit meets every speed, footprint, cost, offline and reproducibility target, measured in RDK-CAL-001 on this repository, on synthetic repositories of up to 400 documents and on a corpus of 38 finished portfolio repositories. It does not meet eight requirements, all of them features still to be written: check coverage (R6), configurable identity (R11), one-command upgrades (R12), the decision-wording guardrail (R13), a getting-started guide (R15), Open Know-How export (R16), ISO A3 sheets (R17) and a single command (R18). Setup time (R1) and macOS and Windows support (R2) cannot be verified at TRL 3 on the machine used.

Requirements R10, R11 and R12 are reworded and R16 to R18 added to reflect the TRL 2 review items adopted for TRL 3 work under Amish's 2026-09-25 instruction, open for his review (RDK-DDR-001). Measured values are from `docs/04-calcs/sizing.py` on a 2-core Linux container with Python 3.11 and kit 1.3.1. Times vary from run to run by up to a factor of two because other jobs share the machine; RDK-CAL-001 gives the ranges.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status at TRL 3 (RDK-CAL-001)	Verification
R1	Setup time from an empty folder to a first passing check	15 min or less on Linux or macOS with Python installed	Not verifiable at TRL 3. The pip install of the kit's dependencies into a fresh environment took 32 to 43 s (1.09 GB), with WeasyPrint's system libraries already present; a new user on a clean machine is still needed	Timed trial by a new user
R2	Platforms	Linux, macOS and Windows (native or WSL2)	Not verifiable at TRL 3: Linux met (this container and CI on Ubuntu); no macOS or Windows machine was available. The TRL 2 note listed Windows native as not met; it is untested rather than shown to fail	CI matrix on three operating systems
R3	Check time per repository	2 s or less	Met: 0.09 s for this repository, 0.10 s median and 0.12 to 0.27 s maximum across the corpus; 0.6 to 0.7 ms per added document, so the 2 s limit is reached at about 2,700 to 3,100 documents	Timed run in CI
R4	PDF render time	10 s or less for three documents	Met: 1.8 to 3.1 s for three documents (0.6 to 1.0 s per document)	Timed run
R5	Concept media render time	60 s or less for a model of up to 20 parts, no GPU	Met: 12 to 23 s for a 20-part model with cutaway, exploded view and scale figure; 5 s for this repository	Timed run on a 20-part model
R6	Check coverage	90% or more of the machine-checkable rules in the standard are checked automatically	Not met: 12 of 20 rules (60%) caught by seeded faults; 18 needed	Seeded faults, one per rule (RDK-CAL-001, section E)
R7	False failures	No check fails a document that follows the standard	Met on the evidence available: 0 failures across 38 conforming TRL 3 repositories and this one	Corpus of conforming repositories
R8	Reproducible output	Same inputs give the same PDF text and images, apart from date and commit	Met: PDF text identical in 5 of 5 documents and PNG renders byte-identical in 4 of 4 across two runs	Render twice and compare
R9	Offline use	Check and render run with no network	Met for check and render with no network interface; <code>viewer.html</code> still loads its viewer script from a CDN	Run with network disabled
R10	Footprint	Kit files in each repository (vendored folder today, pinned stub after D1) 2 MB or less; Python dependencies excluded	Met: 1.05 MB, of which fonts are 94%	<code>du</code> in CI
R11	Configurable identity	Organization name, website, repository owner, author and colors set in <code>readykit.yaml</code>	Not met: 13 hard-coded occurrences of 4 identity values in kit code, style and templates	Render a repository with a second identity

ID	Requirement	Target	Status at TRL 3 (RDK-CAL-001)	Verification
		with Design Molecule defaults; none hard-coded		
R12	Upgrade path	<code>pip install -U readykit</code> followed by <code>readykit upgrade</code> updates a repository to a new kit version and shows the diff	Not met: the kit is copied by hand. All 38 corpus repositories carry kit 1.3.1, yet none has this repository's <code>STANDARDS.md</code> , which shows the drift a versioned package would prevent	Upgrade trial across 5 repositories
R13	Agent guardrails	Check flags TRL above the cap, work beyond the cap, and decisions recorded as made without an owner	Not met (partly): 2 of 3 seeded faults caught; a decision written as made ("Approved") passes	Seeded test documents
R14	Cost to user	\$0; runs on free CI for public repositories	Met: BOM total \$0.00; all 8 direct dependencies free and open source, one of them (CairoSVG) under LGPL-3.0	Dependency license audit
R15	Documentation	Getting-started guide of 5 pages or fewer, plus a reference for every check	Not met: only STANDARDS.md and the kit README exist	Guide review by a new user
R16	Open Know-How export	Writes an <code>okh.yaml</code> manifest (OKH 1.0) with every required field from <code>project.yaml</code> and <code>readykit.yaml</code>	Not met: no exporter; 3 of the 5 required fields are available from <code>project.yaml</code> today, and the author and project link need R11	Validate the manifest against OKH 1.0
R17	Sheet sizes	ANSI B and ISO A3 drawing sheets, chosen per repository	Not met: ANSI B only, fixed in the sheet generator	Render one sheet at each size
R18	Command-line interface	One <code>readykit</code> command with <code>init</code> , <code>check</code> , <code>render</code> , <code>media</code> and <code>upgrade</code> ; the present scripts keep working as wrappers	Not met: three separate scripts	Run each subcommand on a repository

Table 2. Summary of status at TRL 3.

Status	Count	Requirements
Met	8	R3, R4, R5, R7, R8, R9, R10, R14
Not met	8	R6, R11, R12, R13, R15, R16, R17, R18
At risk	0	
Not verifiable at TRL 3	2	R1, R2

Table 3. Requirements not met, and the route adopted for TRL 3 work (RDK-DDR-001) or proposed.

ID	Gap	Route
R6	12 of 20 rules checked	Add at least six of the eight missing checks in Table 4; proposed, awaiting Amish
R11	Hard-coded identity	<code>readykit.yaml</code> with Design Molecule defaults (D2)
R12	Manual kit sync, and drift already present	pip package plus <code>readykit upgrade</code> (D1, D4)
R13	Decision wording unchecked	Promote the portfolio's decision-wording grep into the checker; generic or portfolio-specific wording is open (O4)
R15	No getting-started guide	Write it with the package; proposed, awaiting Amish
R16	No OKH export	Exporter reading <code>project.yaml</code> and <code>readykit.yaml</code> (D6)
R17	ANSI B only	Sheet size as a parameter of the sheet generator, ISO A3 first (D3)
R18	Three scripts	One command with the scripts as wrappers (D4)

Table 4. Machine-checkable rules in STANDARDS.md. "Checked" is the TRL 2 audit; "Seeded fault" is the result of planting one fault per rule in a passing fixture repository (RDK-CAL-001, section E). The two agree on every rule.

Rule	Checked	Seeded fault
Required front matter fields present	Yes	Caught
Document ID matches PRJ-TYP-NNN	Yes	Caught
Document ID unique within the repository	No	Missed
Version format (quoted, n.n)	Yes	Caught
Version equals newest revision row	Yes	Caught
Every revision row complete	Yes	Caught
Status in the allowed set	Yes	Caught
Released documents at 1.0 or later	Yes	Caught
No em dash anywhere in the repository	Partly (controlled documents only), counted as No	Missed in README.md
TRL evidence present for the claimed level	Yes	Caught
TRL and target within the phase cap	Yes	Caught
Required concept media present	Yes (warning at TRL 2)	Caught
Concept media carry the "not for fabrication" label	No	Missed
BOM columns and numbering match the exploded view	No	Missed
BOM fully priced at TRL 3	Yes	Caught
Evidence files listed in project.yaml exist	Yes	Caught
Figures and tables numbered and captioned	No	Missed
Safety note present where hazard keywords appear	No	Missed
README sections in the required order	No	Missed
SI unit formatting (space between value and unit)	No	Missed

Safety: Adding the safety-note check (Table 4) must raise a flag for a person to review. It must never remove the human review of hazards, and a passing check never means a design is safe.